

NAME

codata - library for fundamental physical constants

LIBRARY

codata (**-libcodata**, **-lcodata**)

SYNOPSIS

use *codata*

include "codata.h"

import *pycodata*

DESCRIPTION

codata is a Fortran library providing the fundamental physical constants according to CODATA

List of available constants:

- LATTICE_SPACING_OF_SILICON_2014
- ALPHA_PARTICLE_ELECTRON_MASS_RATIO_2014
- ALPHA_PARTICLE_MASS_2014
- ALPHA_PARTICLE_MASS_ENERGY_EQUIVALENT_2014
- ALPHA_PARTICLE_MASS_ENERGY_EQUIVALENT_IN_MEV_2014
- ALPHA_PARTICLE_MASS_IN_U_2014
- ALPHA_PARTICLE_MOLAR_MASS_2014
- ALPHA_PARTICLE_PROTON_MASS_RATIO_2014
- ANGSTROM_STAR_2014
- ATOMIC_MASS_CONSTANT_2014
- ATOMIC_MASS_CONSTANT_ENERGY_EQUIVALENT_2014
- ATOMIC_MASS_CONSTANT_ENERGY_EQUIVALENT_IN_MEV_2014
- ATOMIC_MASS_UNIT_ELECTRON_VOLT_RELATIONSHIP_2014
- ATOMIC_MASS_UNIT_HARTREE_RELATIONSHIP_2014
- ATOMIC_MASS_UNIT_HERTZ_RELATIONSHIP_2014
- ATOMIC_MASS_UNIT_INVERSE_METER_RELATIONSHIP_2014
- ATOMIC_MASS_UNIT_JOULE_RELATIONSHIP_2014
- ATOMIC_MASS_UNIT_KELVIN_RELATIONSHIP_2014
- ATOMIC_MASS_UNIT_KILOGRAM_RELATIONSHIP_2014
- ATOMIC_UNIT_OF_1ST_HYPERPOLARIZABILITY_2014
- ATOMIC_UNIT_OF_2ND_HYPERPOLARIZABILITY_2014
- ATOMIC_UNIT_OF_ACTION_2014
- ATOMIC_UNIT_OF_CHARGE_2014
- ATOMIC_UNIT_OF_CHARGE_DENSITY_2014
- ATOMIC_UNIT_OF_CURRENT_2014
- ATOMIC_UNIT_OF_ELECTRIC_DIPOLE_MOM_2014

- ATOMIC_UNIT_OF_ELECTRIC_FIELD_2014
- ATOMIC_UNIT_OF_ELECTRIC_FIELD_GRADIENT_2014
- ATOMIC_UNIT_OF_ELECTRIC_POLARIZABILITY_2014
- ATOMIC_UNIT_OF_ELECTRIC_POTENTIAL_2014
- ATOMIC_UNIT_OF_ELECTRIC_QUADRUPOLE_MOM_2014
- ATOMIC_UNIT_OF_ENERGY_2014
- ATOMIC_UNIT_OF_FORCE_2014
- ATOMIC_UNIT_OF_LENGTH_2014
- ATOMIC_UNIT_OF_MAG_DIPOLE_MOM_2014
- ATOMIC_UNIT_OF_MAG_FLUX_DENSITY_2014
- ATOMIC_UNIT_OF_MAGNETIZABILITY_2014
- ATOMIC_UNIT_OF_MASS_2014
- ATOMIC_UNIT_OF_MOMUM_2014
- ATOMIC_UNIT_OF_PERMITTIVITY_2014
- ATOMIC_UNIT_OF_TIME_2014
- ATOMIC_UNIT_OF_VELOCITY_2014
- AVOGADRO_CONSTANT_2014
- BOHR_MAGNETON_2014
- BOHR_MAGNETON_IN_EV_T_2014
- BOHR_MAGNETON_IN_HZ_T_2014
- BOHR_MAGNETON_IN_INVERSE_METERS_PER_TESLA_2014
- BOHR_MAGNETON_IN_K_T_2014
- BOHR_RADIUS_2014
- BOLTZMANN_CONSTANT_2014
- BOLTZMANN_CONSTANT_IN_EV_K_2014
- BOLTZMANN_CONSTANT_IN_HZ_K_2014
- BOLTZMANN_CONSTANT_IN_INVERSE_METERS_PER_KELVIN_2014
- CHARACTERISTIC_IMPEDANCE_OF_VACUUM_2014
- CLASSICAL_ELECTRON_RADIUS_2014
- COMPTON_WAVELENGTH_2014
- COMPTON_WAVELENGTH_OVER_2_PI_2014
- CONDUCTANCE_QUANTUM_2014
- CONVENTIONAL_VALUE_OF_JOSEPHSON_CONSTANT_2014
- CONVENTIONAL_VALUE_OF_VON_KLITZING_CONSTANT_2014
- CU_X_UNIT_2014
- DEUTERON_ELECTRON_MAG_MOM_RATIO_2014
- DEUTERON_ELECTRON_MASS_RATIO_2014
- DEUTERON_G_FACTOR_2014

- DEUTERON_MAG_MOM_2014
- DEUTERON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2014
- DEUTERON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2014
- DEUTERON_MASS_2014
- DEUTERON_MASS_ENERGY_EQUIVALENT_2014
- DEUTERON_MASS_ENERGY_EQUIVALENT_IN_MEV_2014
- DEUTERON_MASS_IN_U_2014
- DEUTERON_MOLAR_MASS_2014
- DEUTERON_NEUTRON_MAG_MOM_RATIO_2014
- DEUTERON_PROTON_MAG_MOM_RATIO_2014
- DEUTERON_PROTON_MASS_RATIO_2014
- DEUTERON_RMS_CHARGE_RADIUS_2014
- ELECTRIC_CONSTANT_2014
- ELECTRON_CHARGE_TO_MASS_QUOTIENT_2014
- ELECTRON_DEUTERON_MAG_MOM_RATIO_2014
- ELECTRON_DEUTERON_MASS_RATIO_2014
- ELECTRON_G_FACTOR_2014
- ELECTRON_GYROMAG_RATIO_2014
- ELECTRON_GYROMAG_RATIO_OVER_2_PI_2014
- ELECTRON_HELION_MASS_RATIO_2014
- ELECTRON_MAG_MOM_2014
- ELECTRON_MAG_MOM_ANOMALY_2014
- ELECTRON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2014
- ELECTRON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2014
- ELECTRON_MASS_2014
- ELECTRON_MASS_ENERGY_EQUIVALENT_2014
- ELECTRON_MASS_ENERGY_EQUIVALENT_IN_MEV_2014
- ELECTRON_MASS_IN_U_2014
- ELECTRON_MOLAR_MASS_2014
- ELECTRON_MUON_MAG_MOM_RATIO_2014
- ELECTRON_MUON_MASS_RATIO_2014
- ELECTRON_NEUTRON_MAG_MOM_RATIO_2014
- ELECTRON_NEUTRON_MASS_RATIO_2014
- ELECTRON_PROTON_MAG_MOM_RATIO_2014
- ELECTRON_PROTON_MASS_RATIO_2014
- ELECTRON_TAU_MASS_RATIO_2014
- ELECTRON_TO_ALPHA_PARTICLE_MASS_RATIO_2014
- ELECTRON_TO_SHIELDED_HELION_MAG_MOM_RATIO_2014

- ELECTRON_TO_SHIELDED_PROTON_MAG_MOM_RATIO_2014
- ELECTRON_TRITON_MASS_RATIO_2014
- ELECTRON_VOLT_2014
- ELECTRON_VOLT_ATOMIC_MASS_UNIT_RELATIONSHIP_2014
- ELECTRON_VOLT_HARTREE_RELATIONSHIP_2014
- ELECTRON_VOLT_HERTZ_RELATIONSHIP_2014
- ELECTRON_VOLT_INVERSE_METER_RELATIONSHIP_2014
- ELECTRON_VOLT_JOULE_RELATIONSHIP_2014
- ELECTRON_VOLT_KELVIN_RELATIONSHIP_2014
- ELECTRON_VOLT_KILOGRAM_RELATIONSHIP_2014
- ELEMENTARY_CHARGE_2014
- ELEMENTARY_CHARGE_OVER_H_2014
- FARADAY_CONSTANT_2014
- FARADAY_CONSTANT_FOR_CONVENTIONAL_ELECTRIC_CURRENT_2014
- FERMI_COUPLING_CONSTANT_2014
- FINE_STRUCTURE_CONSTANT_2014
- FIRST_RADIATION_CONSTANT_2014
- FIRST_RADIATION_CONSTANT_FOR_SPECTRAL_RADIANCE_2014
- HARTREE_ATOMIC_MASS_UNIT_RELATIONSHIP_2014
- HARTREE_ELECTRON_VOLT_RELATIONSHIP_2014
- HARTREE_ENERGY_2014
- HARTREE_ENERGY_IN_EV_2014
- HARTREE_HERTZ_RELATIONSHIP_2014
- HARTREE_INVERSE_METER_RELATIONSHIP_2014
- HARTREE_JOULE_RELATIONSHIP_2014
- HARTREE_KELVIN_RELATIONSHIP_2014
- HARTREE_KILOGRAM_RELATIONSHIP_2014
- HELION_ELECTRON_MASS_RATIO_2014
- HELION_G_FACTOR_2014
- HELION_MAG_MOM_2014
- HELION_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2014
- HELION_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2014
- HELION_MASS_2014
- HELION_MASS_ENERGY_EQUIVALENT_2014
- HELION_MASS_ENERGY_EQUIVALENT_IN_MEV_2014
- HELION_MASS_IN_U_2014
- HELION_MOLAR_MASS_2014
- HELION_PROTON_MASS_RATIO_2014

- HERTZ_ATOMIC_MASS_UNIT_RELATIONSHIP_2014
- HERTZ_ELECTRON_VOLT_RELATIONSHIP_2014
- HERTZ_HARTREE_RELATIONSHIP_2014
- HERTZ_INVERSE_METER_RELATIONSHIP_2014
- HERTZ_JOULE_RELATIONSHIP_2014
- HERTZ_KELVIN_RELATIONSHIP_2014
- HERTZ_KILOGRAM_RELATIONSHIP_2014
- INVERSE_FINE_STRUCTURE_CONSTANT_2014
- INVERSE_METER_ATOMIC_MASS_UNIT_RELATIONSHIP_2014
- INVERSE_METER_ELECTRON_VOLT_RELATIONSHIP_2014
- INVERSE_METER_HARTREE_RELATIONSHIP_2014
- INVERSE_METER_HERTZ_RELATIONSHIP_2014
- INVERSE_METER_JOULE_RELATIONSHIP_2014
- INVERSE_METER_KELVIN_RELATIONSHIP_2014
- INVERSE_METER_KILOGRAM_RELATIONSHIP_2014
- INVERSE_OF_CONDUCTANCE_QUANTUM_2014
- JOSEPHSON_CONSTANT_2014
- JOULE_ATOMIC_MASS_UNIT_RELATIONSHIP_2014
- JOULE_ELECTRON_VOLT_RELATIONSHIP_2014
- JOULE_HARTREE_RELATIONSHIP_2014
- JOULE_HERTZ_RELATIONSHIP_2014
- JOULE_INVERSE_METER_RELATIONSHIP_2014
- JOULE_KELVIN_RELATIONSHIP_2014
- JOULE_KILOGRAM_RELATIONSHIP_2014
- KELVIN_ATOMIC_MASS_UNIT_RELATIONSHIP_2014
- KELVIN_ELECTRON_VOLT_RELATIONSHIP_2014
- KELVIN_HARTREE_RELATIONSHIP_2014
- KELVIN_HERTZ_RELATIONSHIP_2014
- KELVIN_INVERSE_METER_RELATIONSHIP_2014
- KELVIN_JOULE_RELATIONSHIP_2014
- KELVIN_KILOGRAM_RELATIONSHIP_2014
- KILOGRAM_ATOMIC_MASS_UNIT_RELATIONSHIP_2014
- KILOGRAM_ELECTRON_VOLT_RELATIONSHIP_2014
- KILOGRAM_HARTREE_RELATIONSHIP_2014
- KILOGRAM_HERTZ_RELATIONSHIP_2014
- KILOGRAM_INVERSE_METER_RELATIONSHIP_2014
- KILOGRAM_JOULE_RELATIONSHIP_2014
- KILOGRAM_KELVIN_RELATIONSHIP_2014

- LATTICE_PARAMETER_OF_SILICON_2014
- LOSCHMIDT_CONSTANT_273_15_K_100_KPA_2014
- LOSCHMIDT_CONSTANT_273_15_K_101_325_KPA_2014
- MAG_CONSTANT_2014
- MAG_FLUX_QUANTUM_2014
- MOLAR_GAS_CONSTANT_2014
- MOLAR_MASS_CONSTANT_2014
- MOLAR_MASS_OF_CARBON_12_2014
- MOLAR_PLANCK_CONSTANT_2014
- MOLAR_PLANCK_CONSTANT_TIMES_C_2014
- MOLAR_VOLUME_OF_IDEAL_GAS_273_15_K_100_KPA_2014
- MOLAR_VOLUME_OF_IDEAL_GAS_273_15_K_101_325_KPA_2014
- MOLAR_VOLUME_OF_SILICON_2014
- MO_X_UNIT_2014
- MUON_COMPTON_WAVELENGTH_2014
- MUON_COMPTON_WAVELENGTH_OVER_2_PI_2014
- MUON_ELECTRON_MASS_RATIO_2014
- MUON_G_FACTOR_2014
- MUON_MAG_MOM_2014
- MUON_MAG_MOM_ANOMALY_2014
- MUON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2014
- MUON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2014
- MUON_MASS_2014
- MUON_MASS_ENERGY_EQUIVALENT_2014
- MUON_MASS_ENERGY_EQUIVALENT_IN_MEV_2014
- MUON_MASS_IN_U_2014
- MUON_MOLAR_MASS_2014
- MUON_NEUTRON_MASS_RATIO_2014
- MUON_PROTON_MAG_MOM_RATIO_2014
- MUON_PROTON_MASS_RATIO_2014
- MUON_TAU_MASS_RATIO_2014
- NATURAL_UNIT_OF_ACTION_2014
- NATURAL_UNIT_OF_ACTION_IN_EV_S_2014
- NATURAL_UNIT_OF_ENERGY_2014
- NATURAL_UNIT_OF_ENERGY_IN_MEV_2014
- NATURAL_UNIT_OF_LENGTH_2014
- NATURAL_UNIT_OF_MASS_2014
- NATURAL_UNIT_OF_MOMUM_2014

- NATURAL_UNIT_OF_MOMUM_IN_MEV_C_2014
- NATURAL_UNIT_OF_TIME_2014
- NATURAL_UNIT_OF_VELOCITY_2014
- NEUTRON_COMPTON_WAVELENGTH_2014
- NEUTRON_COMPTON_WAVELENGTH_OVER_2_PI_2014
- NEUTRON_ELECTRON_MAG_MOM_RATIO_2014
- NEUTRON_ELECTRON_MASS_RATIO_2014
- NEUTRON_G_FACTOR_2014
- NEUTRON_GYROMAG_RATIO_2014
- NEUTRON_GYROMAG_RATIO_OVER_2_PI_2014
- NEUTRON_MAG_MOM_2014
- NEUTRON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2014
- NEUTRON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2014
- NEUTRON_MASS_2014
- NEUTRON_MASS_ENERGY_EQUIVALENT_2014
- NEUTRON_MASS_ENERGY_EQUIVALENT_IN_MEV_2014
- NEUTRON_MASS_IN_U_2014
- NEUTRON_MOLAR_MASS_2014
- NEUTRON_MUON_MASS_RATIO_2014
- NEUTRON_PROTON_MAG_MOM_RATIO_2014
- NEUTRON_PROTON_MASS_DIFFERENCE_2014
- NEUTRON_PROTON_MASS_DIFFERENCE_ENERGY_EQUIVALENT_2014
- NEUTRON_PROTON_MASS_DIFFERENCE_ENERGY_EQUIVALENT_IN_MEV_2014
- NEUTRON_PROTON_MASS_DIFFERENCE_IN_U_2014
- NEUTRON_PROTON_MASS_RATIO_2014
- NEUTRON_TAU_MASS_RATIO_2014
- NEUTRON_TO_SHIELDED_PROTON_MAG_MOM_RATIO_2014
- NEWTONIAN_CONSTANT_OF_GRAVITATION_2014
- NEWTONIAN_CONSTANT_OF_GRAVITATION_OVER_H_BAR_C_2014
- NUCLEAR_MAGNETON_2014
- NUCLEAR_MAGNETON_IN_EV_T_2014
- NUCLEAR_MAGNETON_IN_INVERSE_METERS_PER_TESLA_2014
- NUCLEAR_MAGNETON_IN_K_T_2014
- NUCLEAR_MAGNETON_IN_MHZ_T_2014
- PLANCK_CONSTANT_2014
- PLANCK_CONSTANT_IN_EV_S_2014
- PLANCK_CONSTANT_OVER_2_PI_2014
- PLANCK_CONSTANT_OVER_2_PI_IN_EV_S_2014

- PLANCK_CONSTANT_OVER_2_PI_TIMES_C_IN_MEV_FM_2014
- PLANCK_LENGTH_2014
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- PLANCK_MASS_ENERGY_EQUIVALENT_IN_GEV_2014
- PLANCK_TEMPERATURE_2014
- PLANCK_TIME_2014
- PROTON_CHARGE_TO_MASS_QUOTIENT_2014
- PROTON_COMPTON_WAVELENGTH_2014
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- PROTON_G_FACTOR_2014
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- PROTON_GYROMAG_RATIO_OVER_2_PI_2014
- PROTON_MAG_MOM_2014
- PROTON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2014
- PROTON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2014
- PROTON_MAG_SHIELDING_CORRECTION_2014
- PROTON_MASS_2014
- PROTON_MASS_ENERGY_EQUIVALENT_2014
- PROTON_MASS_ENERGY_EQUIVALENT_IN_MEV_2014
- PROTON_MASS_IN_U_2014
- PROTON_MOLAR_MASS_2014
- PROTON_MUON_MASS_RATIO_2014
- PROTON_NEUTRON_MAG_MOM_RATIO_2014
- PROTON_NEUTRON_MASS_RATIO_2014
- PROTON_RMS_CHARGE_RADIUS_2014
- PROTON_TAU_MASS_RATIO_2014
- QUANTUM_OF_CIRCULATION_2014
- QUANTUM_OF_CIRCULATION_TIMES_2_2014
- RYDBERG_CONSTANT_2014
- RYDBERG_CONSTANT_TIMES_C_IN_HZ_2014
- RYDBERG_CONSTANT_TIMES_HC_IN_EV_2014
- RYDBERG_CONSTANT_TIMES_HC_IN_J_2014
- SACKUR_TETRODE_CONSTANT_1_K_100_KPA_2014
- SACKUR_TETRODE_CONSTANT_1_K_101_325_KPA_2014
- SECOND_RADIATION_CONSTANT_2014
- SHIELDED_HELION_GYROMAG_RATIO_2014
- SHIELDED_HELION_GYROMAG_RATIO_OVER_2_PI_2014

- SHIELDED_HELION_MAG_MOM_2014
- SHIELDED_HELION_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2014
- SHIELDED_HELION_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2014
- SHIELDED_HELION_TO_PROTON_MAG_MOM_RATIO_2014
- SHIELDED_HELION_TO_SHIELDED_PROTON_MAG_MOM_RATIO_2014
- SHIELDED_PROTON_GYROMAG_RATIO_2014
- SHIELDED_PROTON_GYROMAG_RATIO_OVER_2_PI_2014
- SHIELDED_PROTON_MAG_MOM_2014
- SHIELDED_PROTON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2014
- SHIELDED_PROTON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2014
- SPEED_OF_LIGHT_IN_VACUUM_2014
- STANDARD_ACCELERATION_OF_GRAVITY_2014
- STANDARD_ATMOSPHERE_2014
- STANDARD_STATE_PRESSURE_2014
- STEFAN_BOLTZMANN_CONSTANT_2014
- TAU_COMPTON_WAVELENGTH_2014
- TAU_COMPTON_WAVELENGTH_OVER_2_PI_2014
- TAU_ELECTRON_MASS_RATIO_2014
- TAU_MASS_2014
- TAU_MASS_ENERGY_EQUIVALENT_2014
- TAU_MASS_ENERGY_EQUIVALENT_IN_MEV_2014
- TAU_MASS_IN_U_2014
- TAU_MOLAR_MASS_2014
- TAU_MUON_MASS_RATIO_2014
- TAU_NEUTRON_MASS_RATIO_2014
- TAU_PROTON_MASS_RATIO_2014
- THOMSON_CROSS_SECTION_2014
- TRITON_ELECTRON_MASS_RATIO_2014
- TRITON_G_FACTOR_2014
- TRITON_MAG_MOM_2014
- TRITON_MAG_MOM_TO_BOHR_MAGNETON_RATIO_2014
- TRITON_MAG_MOM_TO_NUCLEAR_MAGNETON_RATIO_2014
- TRITON_MASS_2014
- TRITON_MASS_ENERGY_EQUIVALENT_2014
- TRITON_MASS_ENERGY_EQUIVALENT_IN_MEV_2014
- TRITON_MASS_IN_U_2014
- TRITON_MOLAR_MASS_2014
- TRITON_PROTON_MASS_RATIO_2014

- UNIFIED_ATOMIC_MASS_UNIT_2014
- VON_KLITZING_CONSTANT_2014
- WEAK_MIXING_ANGLE_2014
- WIEN_FREQUENCY_DISPLACEMENT_LAW_CONSTANT_2014
- WIEN_WAVELENGTH_DISPLACEMENT_LAW_CONSTANT_2014

SEE ALSO

codata(3), codata_2010(3), codata_2014(3), codata_2018(3), codata_2022(3)